Intelligent Building Solutions

eBilling MoMo Gateway

Staff User Manual

How to take MTN Mobile Money payments, follow up payments that are stuck, and produce reports from the gateway dashboard.

Dashboard address: `/dashboard/` on the gateway server

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01

Signing in and out

1. Open `/dashboard/` in your browser. You are taken to the sign-in page.
2. Enter your **username** and **password**, then select **Login**. The eye icon shows or hides the password while you type.
3. You land on the **Dashboard**.

Only **staff accounts** can sign in. If you see *"Your username and password didn't match, or the account isn't staff"*, check your details, then ask an administrator to confirm your account has staff access (see For administrators).

To sign out, select the log-out icon next to your name at the top of the sidebar.

02

Finding your way around

Area	What it does
Sidebar (left)	Your name and the sign-out button at the top, then the pages: Dashboard, MoMo Payment, Transactions, Reports, Callback Logs, Applications and Django Admin. The page you are on is highlighted in green.
Toolbar (top)	The panel button hides or shows the sidebar. The breadcrumb (*Home › Transactions › …*) shows where you are; select any part of it to go back.
Page title	The large heading, with the page's main buttons on the right, such as **Export CSV**.

On a phone the sidebar is hidden. Tap the panel button to open it, and tap outside it to close it.

03

Dashboard

The Dashboard gives a quick picture of all payments.

- **Collected**: total of all successful payments.
- **Transactions**: number of payment requests, with how many are still pending.
- **Success rate**: successful payments as a share of payments that have finished (successful + failed). Pending payments are not counted.
- **Applications**: how many systems are registered to request payments.

- **Last 14 days**: bars show requests per day; the line shows the amount collected.
- **By application**: amount collected and success count for each registered system.
- **Recent transactions**: the latest eight payments. Select a row to open it.

New MoMo payment opens the payment page. Export CSV downloads every transaction.

04

Taking a MoMo payment

Use **MoMo Payment** in the sidebar to send a payment request to a customer's phone.

1. Choose the **Application** the payment belongs to, for example `diamond-ebilling`.
2. Enter the customer's **MoMo number**:
   - 10 digits starting with `0`, for example `024 123 4567`, or
   - 12 digits starting with `233`, for example `233241234567`.
   - Spaces are fine. Numbers starting `020`, `027`, `050` or `057` are not MTN numbers and are refused.
3. Enter the **Amount** in cedis.
4. Enter the **Device ID**: the serial number of the meter being paid for.
5. **Username** is filled in with your username. Change it to the customer's account or room name if your team records it that way.
6. Select Send payment request.

The **Status** panel on the right then shows the progress:

Request sent to MTN
MTN accepted the request. The customer's registered MoMo name and number are shown, so you can confirm you have the right person.

Customer approval
The customer gets a prompt on their phone and enters their MoMo PIN. The page checks with MTN every 10 seconds.

Final status

• Payment successful or • Payment failed: declined, wrong PIN, not enough balance, or the prompt expired.

NO ANSWER AFTER 5 MINUTES?

The page stops checking and shows **Still pending**. The payment is not lost: open it from **View transaction** later, or use Check status with MTN.

If you see **Request not sent**, MTN refused the request. The message says why; most often the number is wrong or MTN's service is down. Check the number and try again.

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Transactions

Transactions lists every payment request, newest first, 25 per page.

Finding a payment

- **Search** matches the reference, payer name, payer number, username, device ID or MTN transaction ID. Part of a value is enough, for example `kofi` or `0241`.
- **Status**: Successful, Pending or Failed. **App**: one registered application. Both apply as soon as you change them.
- **From / To**: request dates, both days included.
- Select **Filter** to apply the search and dates. **Clear** removes all filters.

The list

Column	Meaning
Date	When the request was made.
Reference	Our reference for the payment (hover to see it in full). Quote this to MTN or the eBilling team.
Payer	The payer's registered MoMo name with their number underneath. *Name unavailable* means MTN didn't return a name.
App, Username, Device	Which system, account and meter the payment was for.

Column	Meaning
Amount	The amount requested.
Status	● Successful ● Pending ● Failed
Credited	A green tick when the meter has been marked as credited.

Export CSV downloads the list with your current filters applied. The footer shows the total amount of the matching requests.

A single payment

Select a row to open it. **Details** shows everything recorded, including the payer, the MTN transaction ID and MTN's first response. **Lifecycle** shows each stage: request created, sent to MTN, final status, meter credited and eBilling notified.

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Checking payments with MTN

Payments normally update themselves when MTN reports the result. When a payment is stuck on ● Pending, ask MTN directly.

One payment

Open the payment and select **Check status with MTN** at the top right. It only shows while the payment is unfinished. A message tells you the result:

- *MTN reports SUCCESSFUL / FAILED. The transaction has been updated.*
- *MTN reports PENDING. Nothing changed.* The customer hasn't answered yet.
- *MTN has no record of this reference.* The request never reached MTN.

Many payments at once

1. On **Transactions**, filter to the payments you want, for example **Status: Pending** or a date range.
2. A yellow bar shows how many payments in the list are unfinished. Select **Check N with MTN** and confirm.
3. Payments are checked in small batches. A progress bar and counters show how many came back successful, failed, still pending, not found at MTN, or hit an error.

Stop halts after the current batch. Everything already checked is kept.

5 When it finishes, select **Reload list**.

> **MARK ONES MTN HAS NO RECORD OF AS FAILED**
>
> Tick this only when cleaning up old requests. It marks requests that never reached MTN as **Failed** so they stop showing as pending. Leave it unticked for recent payments.

Checking also fills in the payer's number and MoMo name on older payments that don't have them yet.

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Reports

1. Pick the period type: **Monthly**, **Quarterly**, **Half-yearly**, **Yearly** or **Selected period**.
2. Pick the year and month, quarter or half. For **Selected period**, pick the **From** and **To** dates.
3. Optionally pick one **Application**.
4. Select **Run report**.

Period	Covers	Breakdown
Monthly	One calendar month	By day
Quarterly	Q1 Jan–Mar · Q2 Apr–Jun · Q3 Jul–Sep · Q4 Oct–Dec	By month
Half-yearly	H1 Jan–Jun · H2 Jul–Dec	By month
Yearly	1 Jan – 31 Dec	By month
Selected period	Any From–To dates	By day up to about two months, then by month

The report shows **Collected**, **Requests**, **Success rate** and **Average payment** per successful payment. Collected and Requests carry a ▲ / ▼ percentage against the previous period of the same length, for example Q2 against Q1. A dash means the previous period had nothing to compare.

Below that are a trend chart, a **Breakdown** table with a total row (select a day or month to see its transactions) and a **By application** table.

Export CSV downloads the breakdown. **Print** prints a clean copy without the sidebar or filters, stamped with the date and your username.

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Callback Logs

Every message MTN sends to the gateway about a payment is recorded here exactly as received. Use it when investigating a disputed or stuck payment.

- Search by text in the message body (a reference or a number) or by IP address.
- Select a row to see the full message, its headers and where it came from.

If a payment stays pending and there's no matching callback here, MTN hasn't reported the result. Use Check status with MTN.

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Applications

Only **registered applications** can request payments through the gateway. Each has a name, a web address, an IP address and a key.

- **Register**: fill in the form on the right and select **Register**. Leave **Key** blank to generate one, copy it with the clipboard icon and give it to the system's developers.
- **Edit**: change the details and select **Save changes**.
- **Remove**: **Remove app** on the edit page. The application can no longer request payments; its past transactions are kept.

The table also shows how many requests each application has made and how much it has collected.

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What the statuses mean

Status	Meaning	What to do
Pending	Sent to MTN and waiting for the customer, or the result hasn't reached us yet.	Wait a few minutes, then check with MTN.
Successful	The customer approved and the money was collected.	Nothing on the gateway. The meter is credited on the eBilling side.

Status	Meaning	What to do
Failed	Declined, wrong PIN, not enough balance, the prompt expired, or the request never reached MTN.	Ask the customer to try again with a new request.
Credited	The meter has been marked as credited for this payment.	If a successful payment isn't credited, contact the eBilling team with the reference.

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Troubleshooting

Message or problem	Cause and fix
Could not get an access token from MTN	The gateway couldn't sign in to MTN, usually because the MTN credentials are wrong or MTN is down. Tell an administrator.
Enter a valid MTN MoMo number	The number isn't 10 digits starting with 0 or 12 digits starting with 233, or isn't an MTN number.
No registered app found	The application name isn't registered. Add it on Applications.
Payment stuck on Pending	The customer hasn't answered, or MTN's result didn't arrive. Use Check status with MTN.
Payer shows “Name unavailable”	MTN didn't return a name for that number. The payment is not affected.
Payer shows “—”	An older payment with no number recorded. Checking it with MTN fills it in.
Signed out unexpectedly	Your session expired. Sign in again.

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For administrators

Giving someone access

Staff accounts are managed in **Django Admin** (sidebar link, superusers only).

1. Go to **Users › Add user**, set a username and password, and save.

2 On the next screen fill in their name and email, tick **Staff status**, and save.

To remove access, untick **Active**. This keeps their history, which deleting the user would not.

Clearing pending payments automatically

The bulk check can also run on the server:

```
# preview, saves nothing
python manage.py recheck_pending --dry-run
# record results
python manage.py recheck_pending
# also fail requests MTN has no record of
python manage.py recheck_pending --fail-not-found
```

It skips requests less than 10 minutes old (change with `--older-than MINUTES`) and is safe to schedule, for example every 15 minutes with Windows Task Scheduler or cron.

Installation, configuration and the payment API used by eBilling are covered in the project's `README.md`. This manual is also kept in the repository as `docs/STAFF_MANUAL.md`.